Group 2

DO FINTECHS DISCRIMINATE?

December 2nd, 2025

1 Introduction

2 EDA

3 OLS Regressions

4 Diff-in-Diff Regressions

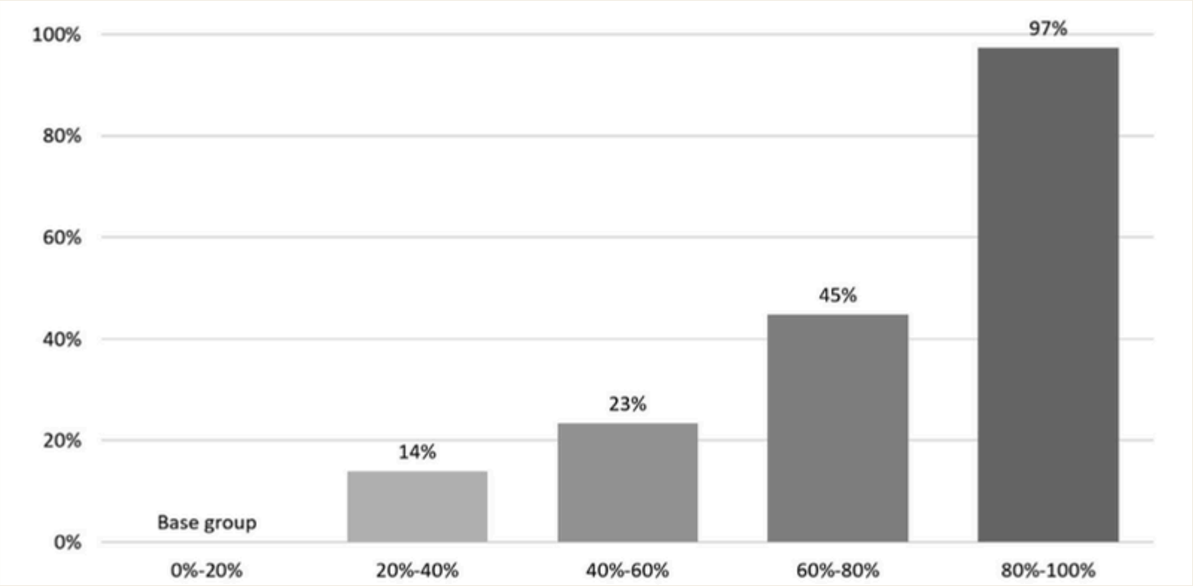
5 Conclusion

BEGLEY AND PURNANDAM (2021)

Color and Credit: Race, Regulation, and the Quality of Financial Services

- Examines how the quality of financial services varies across neighborhoods using **CFPB mortgage-related complaints**
- 1 Standard deviation away leads to:
  - Higher Minority share: +16–17% more complaints (biggest demographic predictor)
  - Lower Income: +8–11% more complaints
  - Education: smaller and often not significant

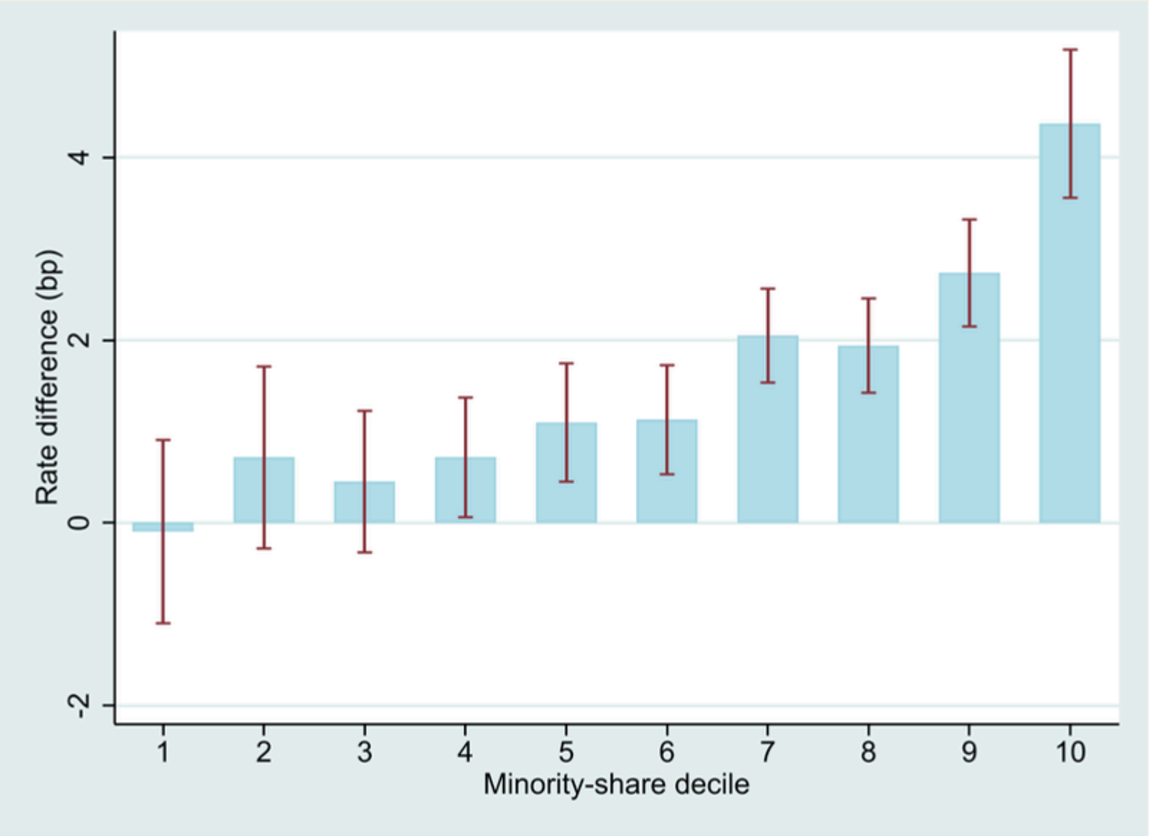
MINORITY SHARE AND COMPLAINT RATES



BARTLETT ET AL (2022)

Consumer-lending discrimination in the FinTech Era

- Test whether Black/Latinx mortgage borrowers pay higher interest rates than white/asian and whether FinTech lenders are more or less discriminatory than traditional lenders.
- Rate gaps rise sharply in high-minority neighborhoods, showing discrimination is strongest in minority-dense areas.
- Algorithmic lending can reduce discrimination relative to face-to-face lenders



1 Introduction

2 EDA

3 OLS Regressions

4 Diff-in-Diff Regressions

5 Conclusion

DATA SOURCES

US Complaints and Demographics

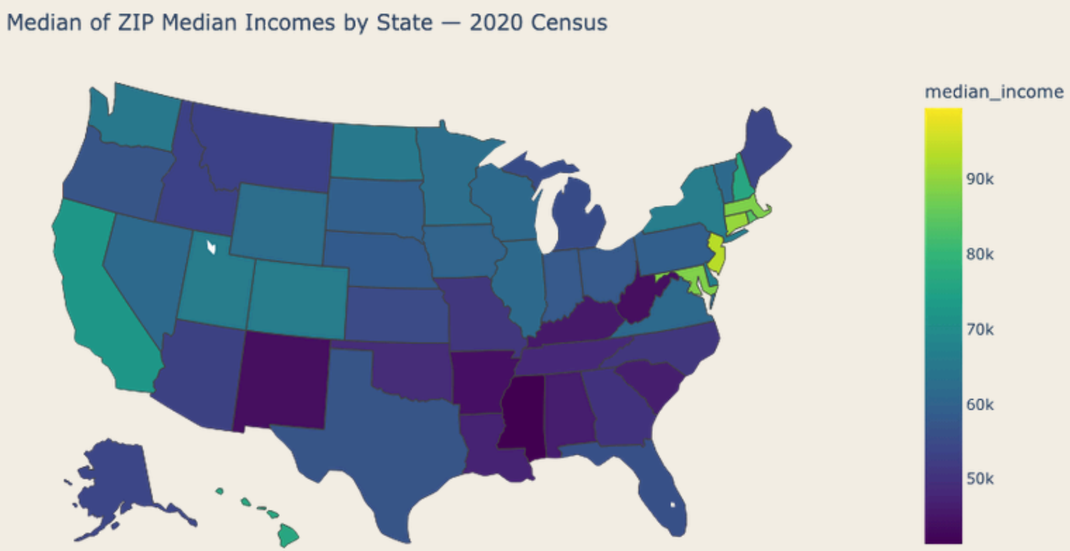
- A public dataset of 11.7 million consumer complaints (2011–2025)
  - Issue (company, product, issue)
  - Complaint (submission channel, resolution, date, narrative)
  - Location (zip code)

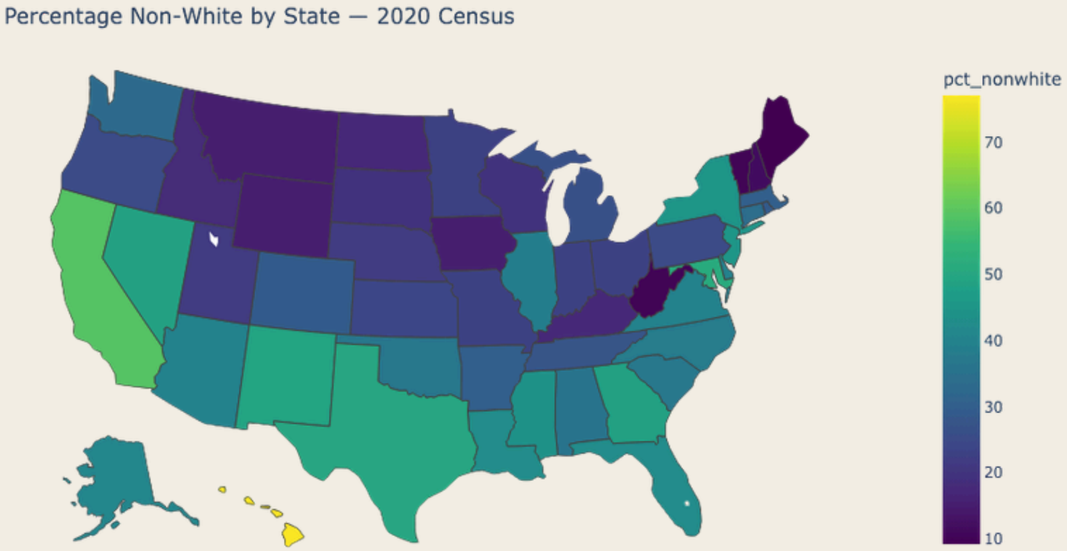
Zip Code

- P3 - Population by race
- S1901 - Household income (mean, median income in last 12 months)
- S1501 - Education attainment (percent with high school, college, graduate degree)

US CENSUS INSIGHTS

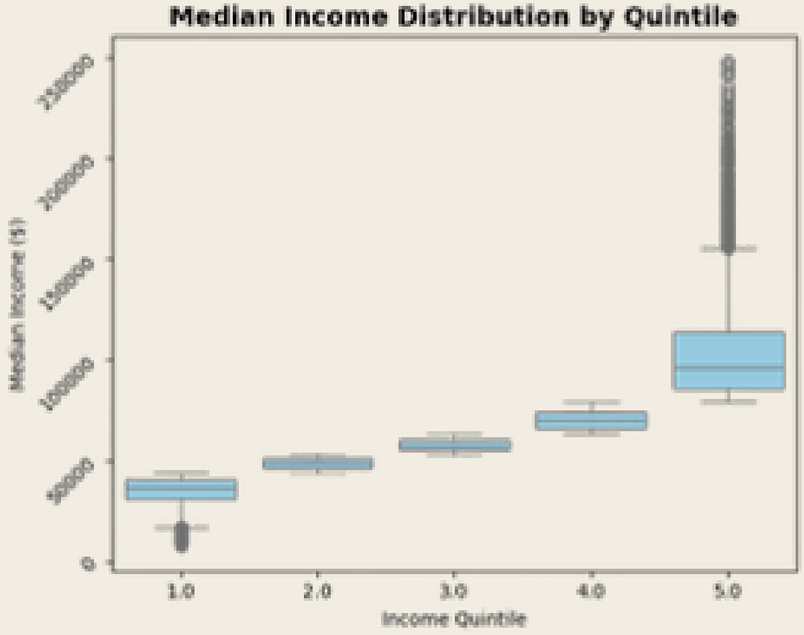
Income and Ethnicity by State

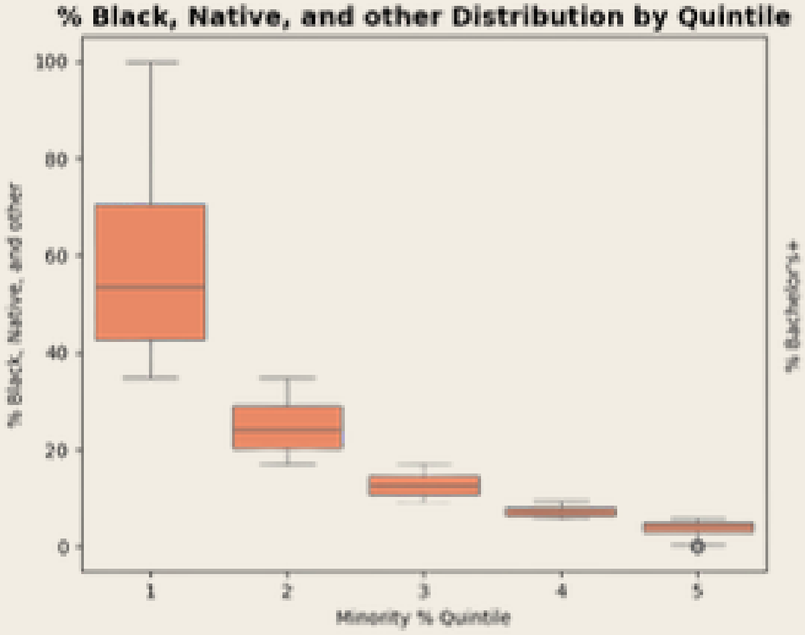


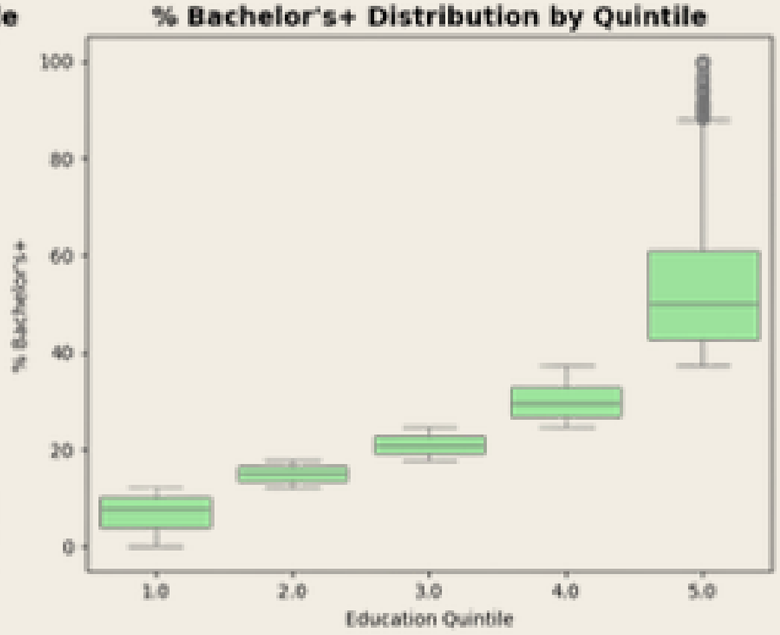


US CENSUS INSIGHTS

Income, Ethnicity, Education Quintiles

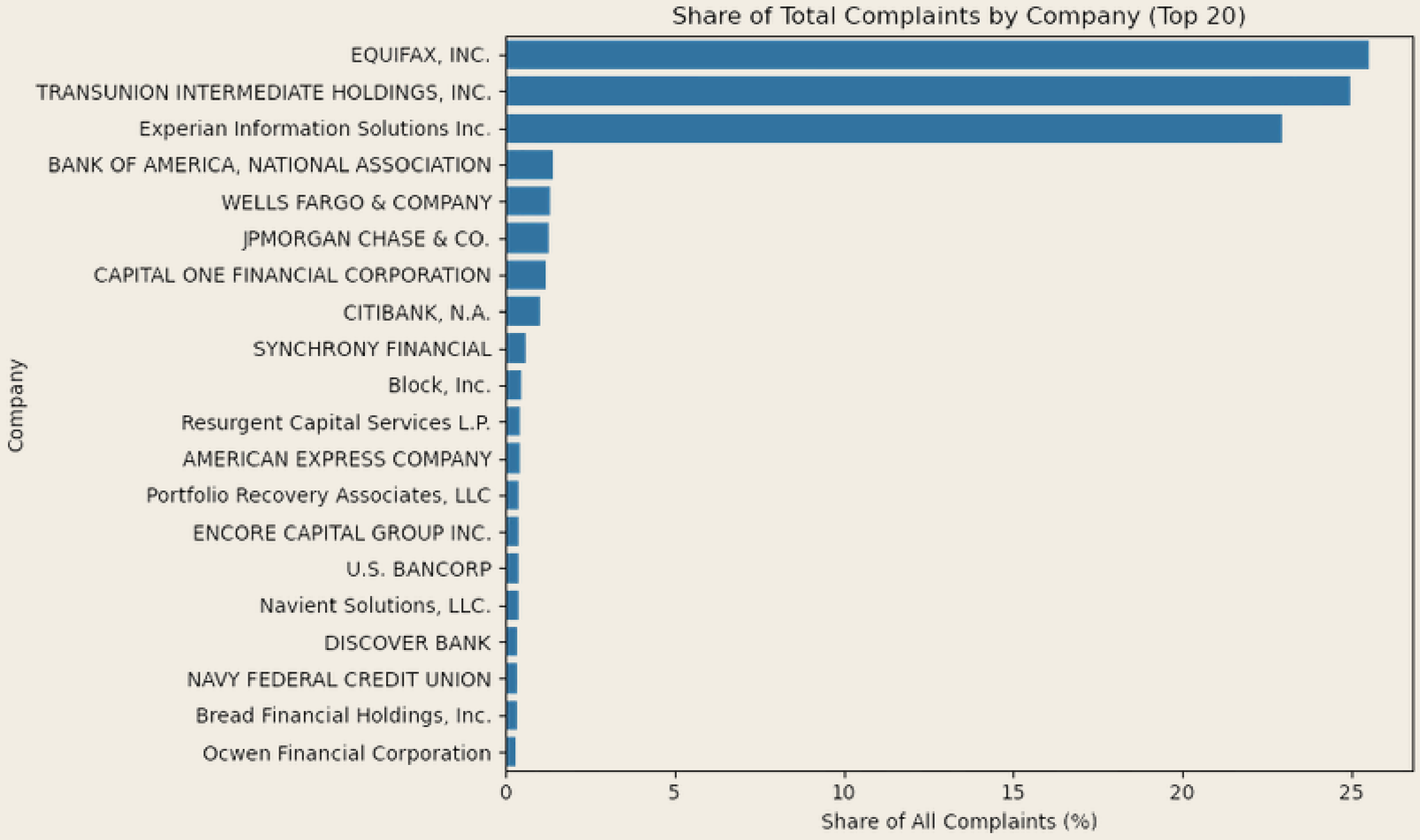






CFPB INSIGHTS

Complaints by Company



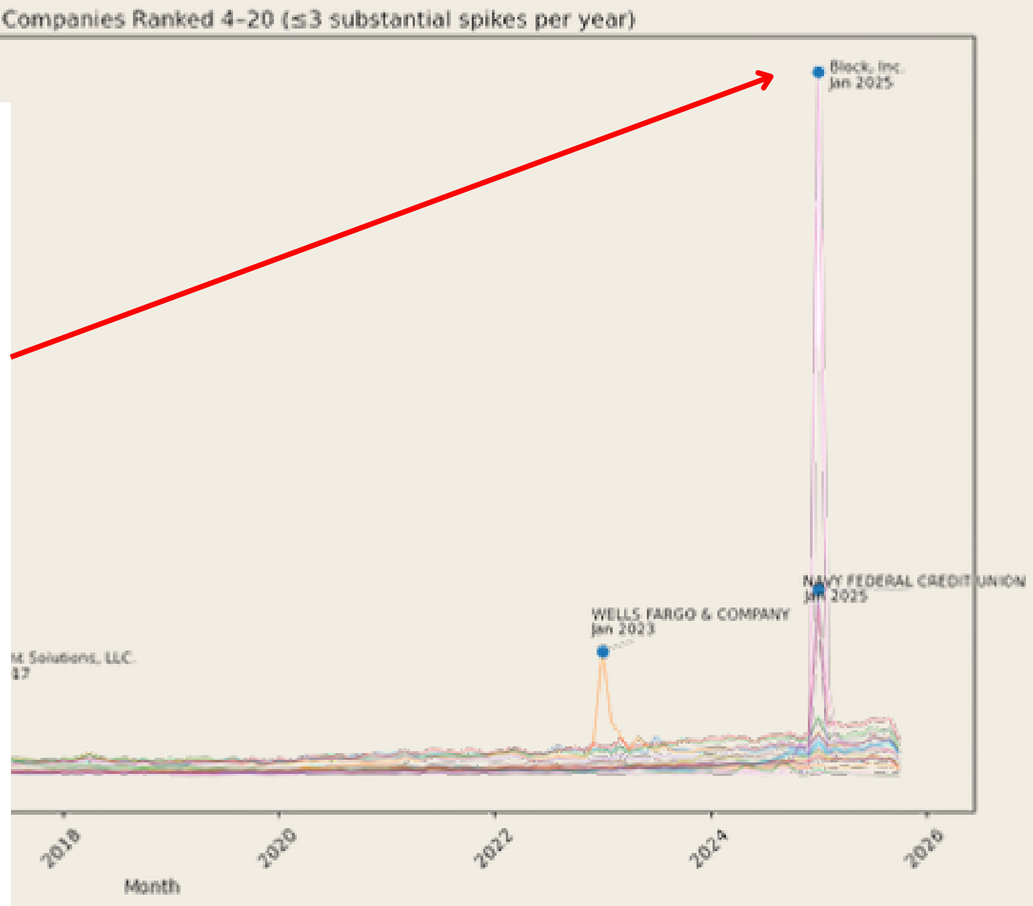
CFPB INSIGHTS

Non Credit Union Complatins over time

/ Enforcement Actions

Block, Inc.

On January 16, 2025, the Bureau issued an order against Block, Inc. (Block), a nonbank headquartered in Oakland, California which operates Cash App, a mobile payments application through which consumers can send and receive money through peer-to-peer transfers. The Bureau found that for years after the inception of Cash App, Block failed to provide effective customer service for Cash App, including by failing to provide live telephone agents, which prevented consumers from being able to have their financial issues addressed in a proper and timely fashion and resulted in fake customer service lines through which consumers' information would be stolen, in a manner that was unfair in violation of the Consumer Financial Protection Act of 2010 (CFPA). The Bureau also found that Block failed to take timely, appropriate, and effective measures to prevent, detect, limit, and address fraud on the Cash App platform in a manner that was unfair in violation of the CFPA. The Bureau further found that Block used the card network chargeback process as a substitute for fulfilling its obligations under the Electronic Fund Transfer Act (EFTA) and Regulation E to investigate and resolve disputes about unauthorized transactions in a timely manner in violation of the CFPA's prohibition on unfair practices. In addition, the Bureau found that Block engaged in deception by misrepresenting that it protected consumers from unauthorized transfers and had a telephone line to report such unauthorized transfers. The Bureau also found that Block failed to comply in multiple ways with the requirements of EFTA and Regulation E, including regarding error resolution. The order requires Block to change its practices to comply with the law, pay at least $75 million in redress to consumers, up to $120 million. The order further requires Block to pay $55 million in civil money penalties.



CFPB INSIGHTS

Complaint Narratives

- 29% of complaints include a narrative
- Largely negative - aligned with nature of complaints
- Topic modelling via BERT and LDA - out of scope for this research question

1 Introduction

2 EDA

3 OLS Regressions

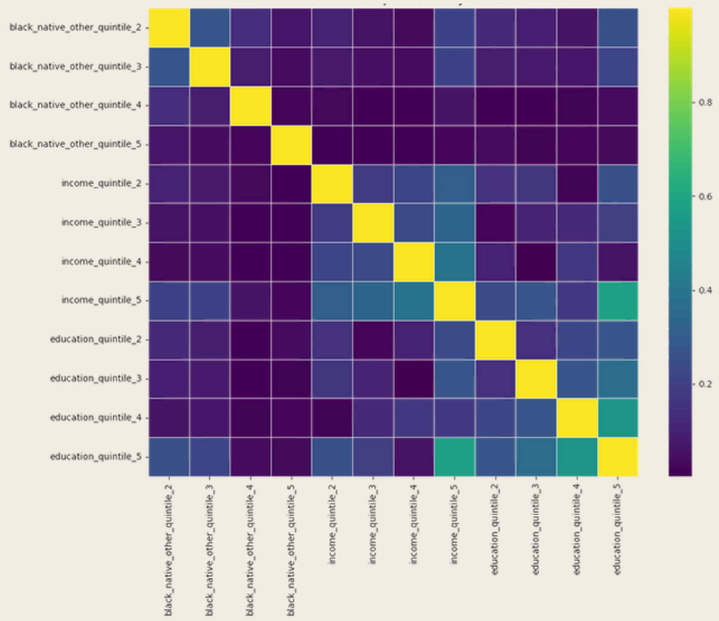
4 Diff-in-Diff Regressions

5 Conclusion

CORRELATIONS

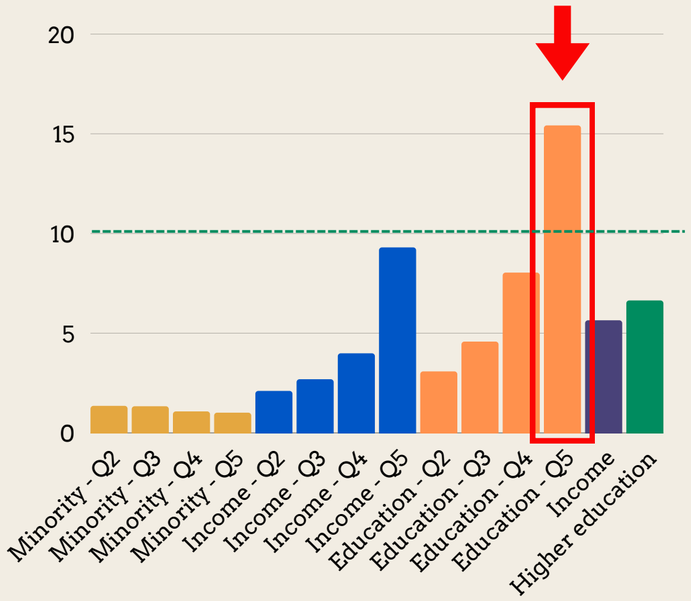
Tests: Cramer's V, Variance Inflation Factor (VIF)

PAIRWISE CORRELATIONS: CRAMER'S V



- **High Correlation:**
  - highest income and education quintiles
  - within high education quintiles

VARIANCE INFLATION FACTOR



- Most variables with **VIF lower than 10**
- **The most collinear variable:** Q5 education quintile

MODEL SPECIFICATION

Baseline: non-fintech company, high minority, lowest income and education ZIP codes

Model Formulation:

$$\log(\text{norm_num_complaints}_{ict}) = \mathbf{Q}_c^{\text{minority}}\boldsymbol{\beta}_m + \mathbf{Q}_c^{\text{income}}\boldsymbol{\beta}_i + \mathbf{Q}_c^{\text{education}}\boldsymbol{\beta}_e + \left(\text{Fintech} \cdot \mathbf{Q}_c^{\text{minority}}\right)\boldsymbol{\gamma} + \mathbf{Z}_c\boldsymbol{\theta} + \lambda_i + \tau t + \varepsilon_{ict}$$

1. MINORITY | **2. INCOME** | **3. EDUCATION** | **4. ALL VARIABLES**

- a. Quintiles
- b, With interaction
- c. With controls

MODELLING RESULTS

Baseline: non-fintech company, high minority, lowest income and education ZIP codes

	MINORITY			INCOME			EDUCATION			ALL VARIABLES		
	(1a)	(1b)	(1c)	(2a)	(2b)	(2c)	(3a)	(3b)	(3c)	(4a)	(4b)	(4c)
Intercept	0.0407***	0.0428***	0.0611***	0.0675***	0.0627***	0.0615***	0.0983***	0.0924***	0.0908***	0.0883***	0.0901***	0.0916***
black_native_other_quintile_2[T.True]	0.0052***	0.0058***	0.0157***							0.0162***	0.0167***	0.0170***
black_native_other_quintile_3[T.True]	0.0203***	0.0214***	0.0345***							0.0323***	0.0333***	0.0340***
black_native_other_quintile_4[T.True]	0.0770***	0.0791***	0.0871***							0.0854***	0.0873***	0.0876***
black_native_other_quintile_5[T.True]	0.2544***	0.2563***	0.2562***							0.2525***	0.2545***	0.2545***
time_trend	-0.0000***	-0.0000***	-0.0000***	-0.0001***	-0.0001***	-0.0001***	-0.0001***	-0.0001***	-0.0001***	-0.0000***	-0.0000***	-0.0000***
fintech_x_qm_2		-0.0043**	-0.0040**		0.0069***	0.0092***		0.0090***	0.0087***		-0.0036**	-0.0035**
fintech_x_qm_3		-0.0087***	-0.0082***		0.0185***	0.0209***		0.0205***	0.0200***		-0.0079***	-0.0078***
fintech_x_qm_4		-0.0191***	-0.0187***		0.0646***	0.0644***		0.0648***	0.0642***		-0.0184***	-0.0184***
fintech_x_qm_5		-0.0192**	-0.0197**		0.2381***	0.2361***		0.2321***	0.2318***		-0.0198**	-0.0198**
income_zip_10k			-0.0010***			0.0023***			0.0004***			-0.0002*
share_bach_or_higher_zip			-0.0526***			-0.0483***			-0.0011			-0.0123***
income_quintile_2[T.True]				-0.0101***	-0.0105***	-0.0108***				-0.0058***	-0.0057***	-0.0056***
income_quintile_3[T.True]				-0.0150***	-0.0156***	-0.0158***				-0.0084***	-0.0083***	-0.0080***
income_quintile_4[T.True]				-0.0183***	-0.0190***	-0.0195***				-0.0099***	-0.0099***	-0.0093***
income_quintile_5[T.True]				-0.0208***	-0.0219***	-0.0223***				-0.0133***	-0.0133***	-0.0116***
education_quintile_2[T.True]							-0.0286***	-0.0288***	-0.0291***	-0.0287***	-0.0287***	-0.0280***
education_quintile_3[T.True]							-0.0420***	-0.0422***	-0.0427***	-0.0402***	-0.0401***	-0.0388***
education_quintile_4[T.True]							-0.0515***	-0.0516***	-0.0525***	-0.0471***	-0.0471***	-0.0447***
education_quintile_5[T.True]							-0.0522***	-0.0530***	-0.0547***	-0.0526***	-0.0526***	-0.0476***
R-squared	0.1096	0.1099	0.1235	0.0113	0.0195	0.0236	0.0250	0.0330	0.0331	0.1342	0.1345	0.1346
R-squared Adj.	0.1095	0.1098	0.1234	0.0112	0.0194	0.0236	0.0250	0.0329	0.0330	0.1342	0.1344	0.1346
Adj. R^2	0.1095	0.1098	0.1234	0.0112	0.0194	0.0236	0.0250	0.0329	0.0330	0.1342	0.1344	0.1346
N	1,701,381	1,701,381	1,701,381	1,701,381	1,701,381	1,701,381	1,701,381	1,701,381	1,701,381	1,701,381	1,701,381	1,701,381
R^2	0.1096	0.1099	0.1235	0.0113	0.0195	0.0236	0.0250	0.0330	0.0331	0.1342	0.1345	0.1346

Standard errors in parentheses.
* p<.1, ** p<.05, ***p<.01

non-fintechs:

ZIP codes with **low minority share** file substantially **more complaints**

1

fintechs vs non-fintechs:

Fintech companies have **fewer complaints** relative to non-fintechs

4

all institutions:

Higher-income ZIP codes file **fewer complaints**

2

all institutions:

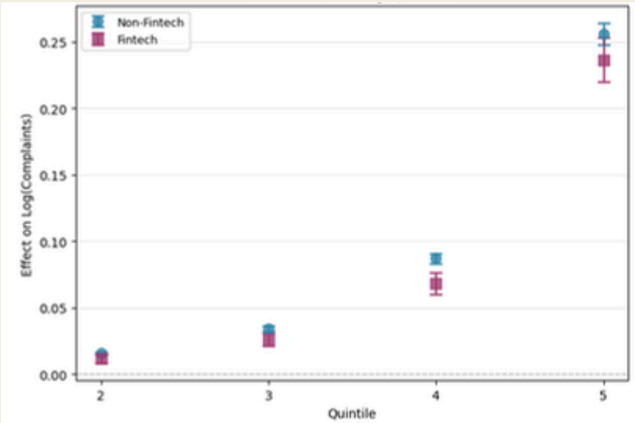
Higher-education ZIPs file **fewer complaints**

3

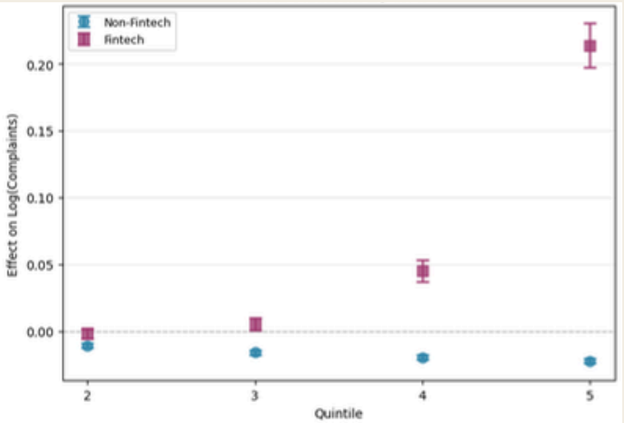
FINTECH VS NON-FINTECH

How does the **number of complaints** differ across **different quintiles**?

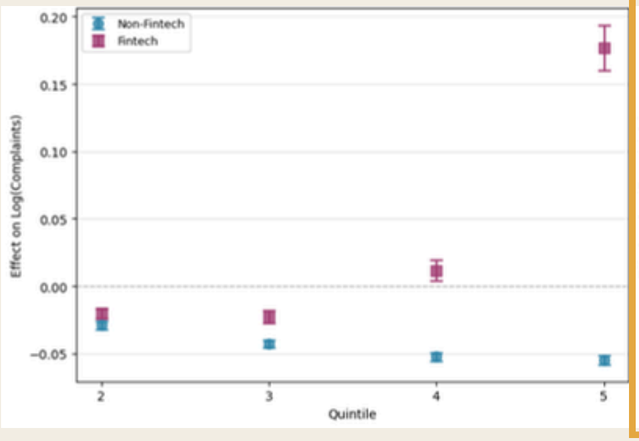
MINORITY



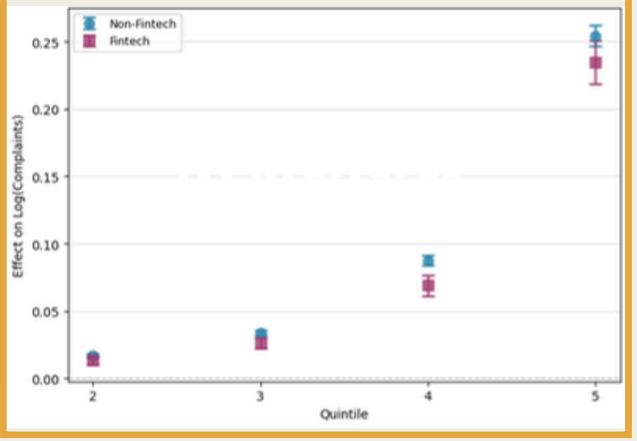
INCOME



EDUCATION



ALL VARIABLES



high-minority share <--> low minority share

LIMITATIONS

And potential caveats...

1

Manual Labelling of the Institutions - fintech vs non-fintech

- **Misclassification** or subjective judgment may affect the accuracy of the comparisons
- Not all companies are banks

2

ZIP codes as a proxy for minority

- Individual-level characteristics are not available in the complaint data
- This might **introduce additional noise to the analysis**

1 Introduction

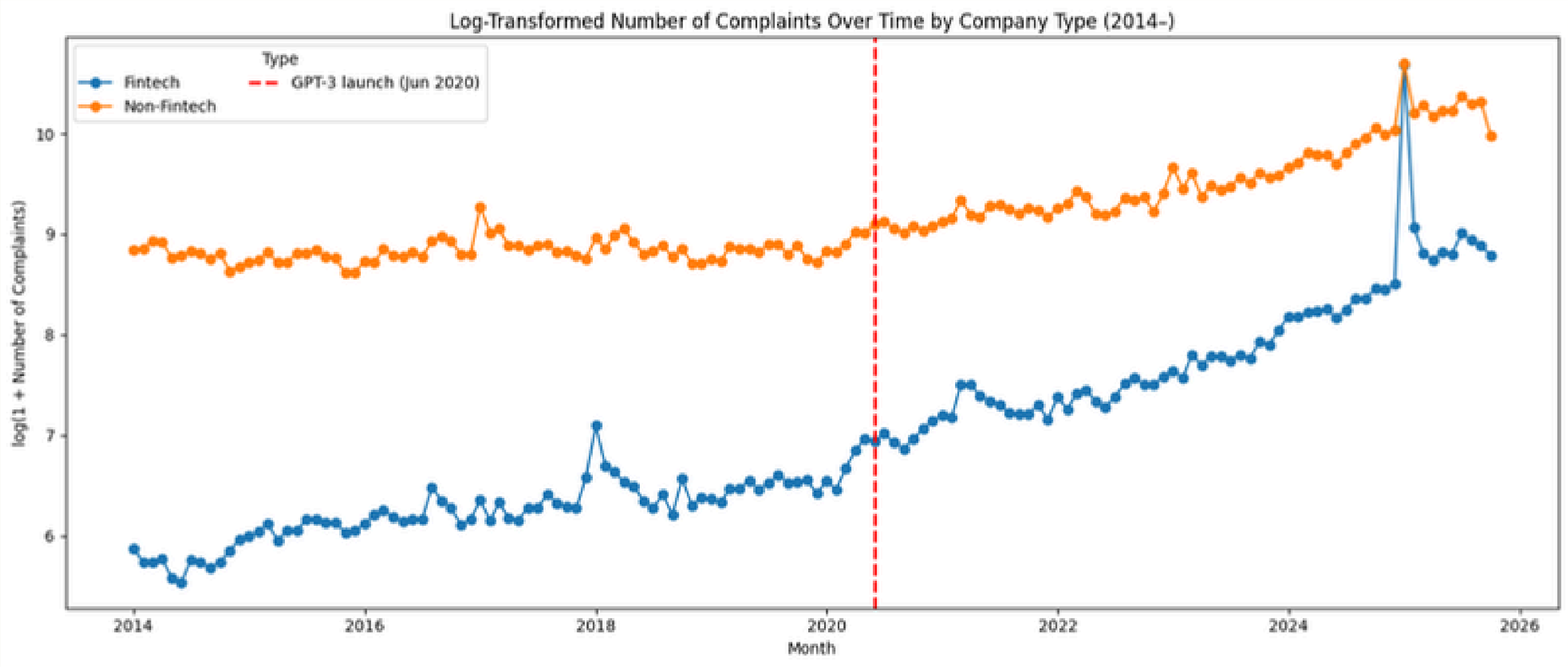
2 EDA

3 OLS Regressions

4 Diff-in-Diff regressions

5 Conclusion

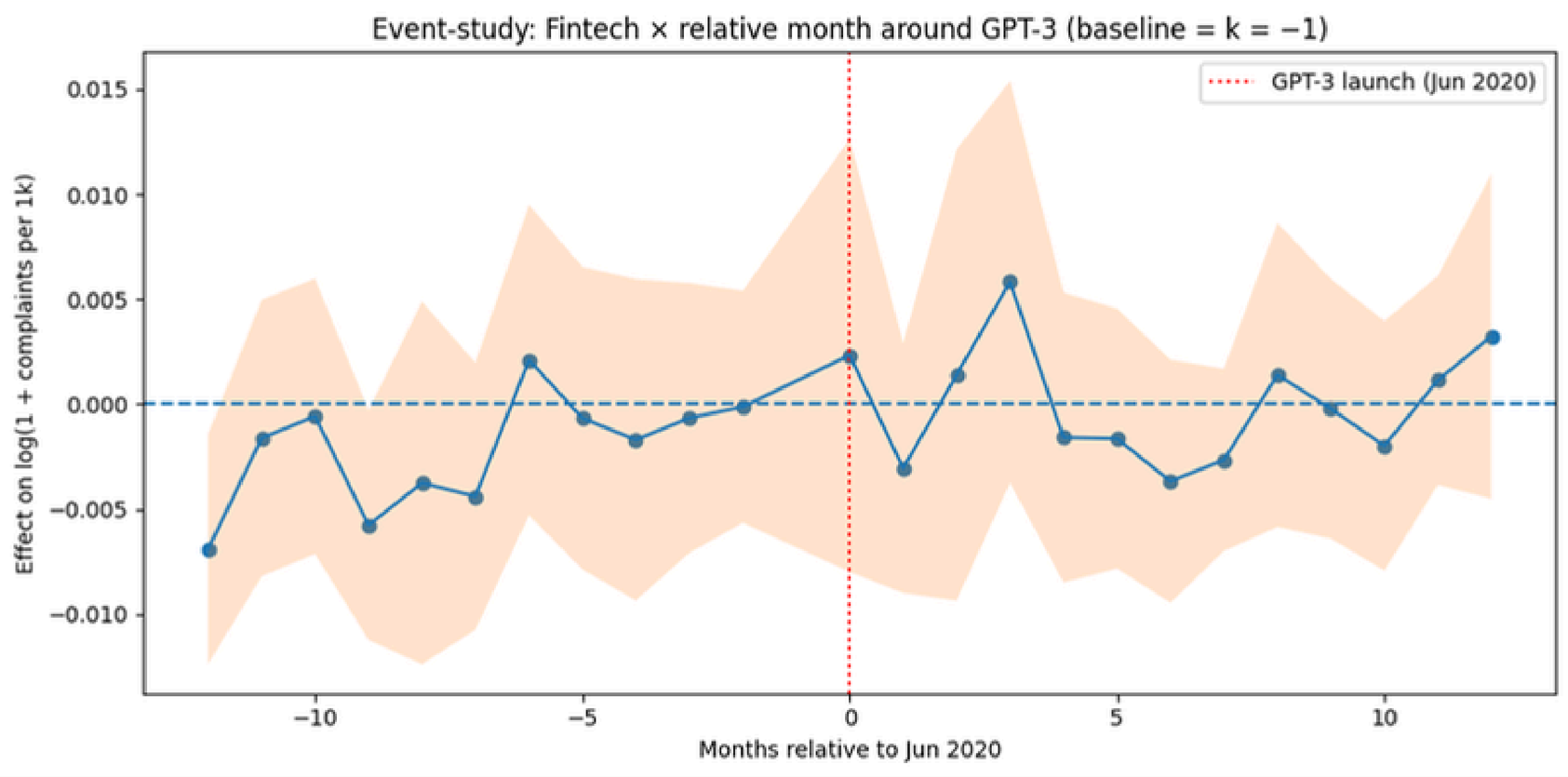
DIFF-IN-DIFF: GPT-3 SHOCK (JUN '20)



Baseline model: $\log(\text{norm_complaints}) = \beta_0 + \text{FE}_{\text{months}} + \text{FE}_{\text{banks}} + \beta_1(\text{Fintech} \times \text{Post})$

$$\beta_1(\text{Fintech} \times \text{Post}) = 0.0017 \quad (p = 0.200)$$

EVENT STUDY: GPT-3 SHOCK (JUN ‘20)



No post event divergence

1 Introduction

2 EDA

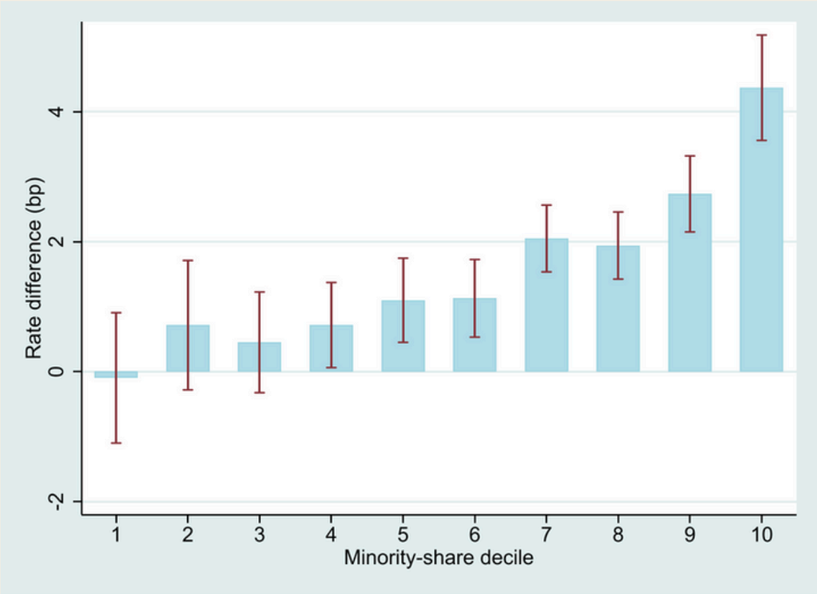
3 OLS Regressions

4 Diff-in-Diff regressions

5 Conclusion

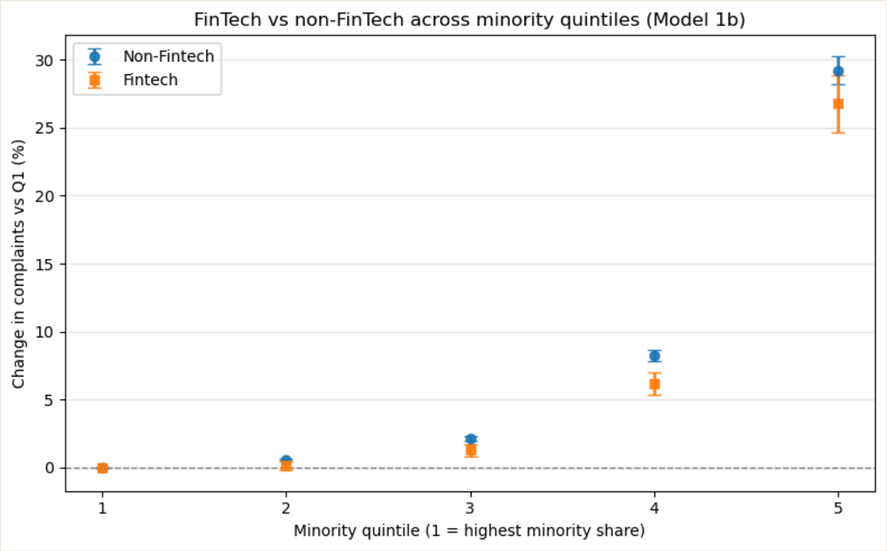
RESULTS

BARTLETT ET AL. (2022)



FinTech slightly reduces pricing discrimination, and high minority geographies still face the worst outcomes

GROUP 2 (2025)



FinTech reduces overall complaints, but this benefit weakens in the highest-minority

Group 2

Thank you for the attention

Q&A

December 2nd, 2025